

Keyur Parag Joshi

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Current Work Experience

Google – Cloud AI – Software Engineer L4

August 2024 — Ongoing

- Designed & implemented infrastructure for compartmentalized testing of AI agents that modify the execution environment *used to validate candidates for Gemini models from Gemini 2.5 onwards*
 - Implemented mock tools that model real services for sandboxed AI agent evaluation *used to evaluate Gemini models for Gemini Enterprise*
 - Onboarded evaluation benchmarks for AI agents in key domains (e.g., enterprise AI and customer service) *used to evaluate Gemini models for Gemini Enterprise and to specialize Gemini for a major customer*
 - Created and improved tools for analyzing agent evaluation results, especially failure cases *used to diagnose multiple bugs and performance regressions and validate candidates for Gemini models*
 - Integrated dynamic user simulation for the testing of conversational AI agents into the Google Agent Development Kit (ADK) *enabled evaluation of realistic, multi-turn user conversations (e.g., for chatbots) within ADK*
 - Integrated evaluation-guided automatic AI agent optimization into the Google Agent Development Kit (ADK) *enabled optimization of instructions and skills of ADK-based agents*
 - Fixed multiple blocking issues, improved code quality, and added QoL features across the agent quality testing infra
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Previous Work Experience

Microsoft Research India – Research Intern

May 2022 — August 2022

- Designed and implemented an [open-source library](#) for training neural networks such as RNNs using only fixed-point arithmetic
 - Demonstrated proof-of-concept that achieved accuracy similar to that of a floating-point implementation of the same RNN
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Education

University of Illinois Urbana-Champaign (UIUC), USA

August 2017 — May 2024

PhD in Computer Science

Research focus – analysis of quantitative uncertainty in programs and systems; effective application of approximations

Indian Institute of Technology, Hyderabad (IITH), India

August 2013 — May 2017

Bachelor of Technology (Honours) in Computer Science and Engineering

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Skills

Technical:

- Experienced: Python, Mercurial, Git, Markdown, Google Workspace, Microsoft Office
- Familiar: Bazel, Jujutsu, C/C++, $\LaTeX$, Bash, LLVM, PyTorch, Go, Java, Make
- Learning: Rust, Docker

Soft Skills: Teamwork (10+ team members), Communication, Documentation (developer and end-user), Leadership

Languages: English (native proficiency), German (CEFR A2+)

Open-Source Tools

Automated Agent Optimization in Google Agent Development Kit: *Evaluation-driven optimization of AI agents*

- Contributions: feature [implementation](#), testing, and [user documentation](#)

User Simulation in Google Agent Development Kit: *Dynamic, model-driven user simulation for testing AI agents*

- Contributions: feature [implementation](#), testing, [user documentation](#), and editing of the announcement [blog post](#)

Fixed-Point-RNN-Training: *Tool for training RNNs using exclusively fixed-point arithmetic*

- Tool, example, tutorial: github.com/KPJoshi/Fixed-Point-RNN-Training

AxProf: *Tool for statistical analysis of the precision of approximate algorithms*

- Successfully found bugs in multiple approximate algorithm implementations
- Tool, examples, tutorial: axprof.org

Selected Publications

- **FastFlip: Compositional SDC Resiliency Analysis**
Keyur Joshi, Rahul Singh, Tommaso Bassetto, Sarita Adve, Darko Marinov, Sasa Misailovic
International Symposium on Code Generation and Optimization (CGO 2025)
 - **Compositional Analysis of the Effects of Uncertainty on Computations**
Keyur Joshi
PhD Dissertation (2024)
 - **GAS: Generating Fast and Accurate Surrogate Models for Autonomous Vehicle Systems**
Keyur Joshi, Chiao Hsieh, Sayan Mitra, Sasa Misailovic
International Symposium on Software Reliability Engineering (ISSRE 2024)
 - **Diamont: Dynamic Monitoring of Uncertainty for Distributed Asynchronous Programs**
Vimuth Fernando, Keyur Joshi, Jacob Laurel, Sasa Misailovic
International Conference on Runtime Verification (RV 2021)
 - **Aloe: Verifying Reliability of Approximate Programs in the Presence of Recovery Mechanisms**
Keyur Joshi, Vimuth Fernando, Sasa Misailovic
IEEE/ACM International Symposium on Code Generation and Optimization (CGO 2020)
 - **Statistical Algorithmic Profiling for Randomized Approximate Programs**
Keyur Joshi, Vimuth Fernando, Sasa Misailovic
ACM/IEEE International Conference on Software Engineering (ICSE 2019)
 - **Verifying Safety and Accuracy of Approximate Parallel Programs via Canonical Sequentialization**
Vimuth Fernando, Keyur Joshi, Sasa Misailovic
ACM SIGPLAN Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA 2019)
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